

RESEARCH ARTICLE

Synthetic Social Engineering Scenario Generation Using LLMs for Awareness-Based Attack Resilience

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ABSTRACT Social engineering is found in a strong majority of cyberattacks today, as it is a powerful manipulation tactic that does not require the technical skills of hacking. Calculated social engineers utilize simple communication to deceive and exploit their victims, all by capitalizing on the vulnerabilities of human nature: trust and fear. When successful, this inconspicuous technique can lead to millions of dollars in losses. Social engineering is not a one-dimensional technique; criminals often leverage a combination of strategies to craft a robust yet subtle attack. In addition, offenders are continually evolving their methods in efforts to surpass preventive measures. A common utility to defend against social engineering attacks is detection-based software. Security awareness, however, is a valuable approach that is often eclipsed by automated tech solutions. Awareness establishes a strong first line of defense against these ever-changing attacks. This study utilizes four data-supplemented large language models to generate custom social engineering scenarios with the goal of supporting strong example-driven security awareness programs. The performances of BERT, T5, GPT-3.5, and Llama 3.1 are comparatively analyzed, with Llama 3.1 producing the highest quality scenarios based on multiple metrics, including LLM-as-a-judge. Through chain-of-thought prompting, the Llama 3.1: Interactive version is capable of generating superiorly realistic, detailed, and specific attack scenarios, which are invaluable to improving hands-on curriculums and security exercises.

INDEX TERMS Fine-tuning, LLM, phishing, prompt engineering, scenario generation, social engineering.

I. INTRODUCTION

Social engineering is a malicious tactic that involves the psychological manipulation of victims to achieve goals like theft or access. The malevolent party, often a cybercriminal, uses careful communication to take advantage of their target. Although social engineering methods may involve technological aspects such as malware or spoofing, the initial exploit relies on deceptive contact. Attacks often involve fabrications that apply pressure and prey on human weaknesses, such as the inclination to help someone in need. Elements of urgency, distress, sympathy, and even flattery covertly wear down the victim's guard and influence them to comply. Recipients of social engineering attacks range from

individuals to entire companies, and although most attacks are unsuccessful, a single mistake can result in devastating consequences, from major financial losses to massive data breaches. Phishing, pretexting, baiting, and business email compromise (BEC) are some of the most prominent methods used in social engineering attacks, and new techniques are being discovered every year.

Social engineers' approaches are becoming increasingly intricate, subtle, and multi-layered, which makes them more difficult to detect by current counteractive solutions. Two widely used practices are artificial intelligence (AI)-led attack detection software and multi-factor authentication (MFA). Although these methods are shown to have a positive impact, they are not foolproof. In addition, even though powerful, specialized, and contemporary AI services may be leveraged by large organizations, they are neither

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practical nor accessible for everyday scenarios. Therefore, it is imperative to emphasize security awareness training as an essential factor in preventing successful social engineering attacks. Humans are the first point of contact in a social engineering attack, and well-developed training fosters vital skills such as online vigilance, threat recognition, and risk avoidance. Currently, online security awareness programs are mandated by select employers, which leaves the general population without proper training. Furthermore, many company-led awareness programs are noncomprehensive, surface-level, and only provided to a subset of employees.

Recent advancements in deep learning, generative AI, and natural language processing (NLP) have given rise to an assortment of powerful large language models (LLMs) capable of producing text that rivals human writing. As demonstrated by several benchmarks, state-of-the-art LLMs such as Llama 3.1 display superior strength in conversational, creative, and domain-specific language applications. Therefore, the goal of this study is to develop a model that is capable of creating high-quality social engineering scenarios to supplement in-depth, accessible, and impactful security awareness curriculums, with an emphasis on attack simulation. Four LLMs of varying architecture, BERT, T5, GPT-3.5, and Llama 3.1, will be tuned for this task, and their performances will be compared using style and evaluation metrics to determine which model is the most suitable for scenario generation.

The contributions of this paper to the body of knowledge are as follows:

- Provide a detailed background on social engineering, including recent major attacks and contemporary attack trends.
- Analyze social engineering threat statistics to highlight its prevalence and danger, discuss the efficacy of defense strategies (both human- and tech-based), and discuss the insufficiency of existing cybersecurity awareness programs.
- Highlight the lack of existing studies on social engineering attack generation and review works involving text generation in other domains.
- Compare the performance of four LLMs of different complexities in the task of dataset-supplemented social engineering scenario generation.
- Demonstrate the value of combining multiple standard evaluation techniques for a robust assessment of open-ended generated text.
- Emphasize the potential of utilizing our strongest model configuration to automate the development of high-quality and realistic social engineering scenarios for improved simulation-based awareness curriculums.

II. BACKGROUND

This section will describe social engineering techniques, major attacks, threat trends, and notable statistics.

A. SOCIAL ENGINEERING

Social engineering is an intricate topic that involves an array of goals, tactics, and attack types.

Social engineering is a potent method of deception found in approximately 70-90% of all reported cyberattacks [1]. Social engineering is the process of using premeditated and deceitful social contact to psychologically influence victims [2], [3]. Social engineers strive to manipulate their target into following instructions that align with an end goal [1], [2], [3]. Common goals of social engineering operations include obtaining sensitive information, gaining restricted access, spreading malware, stealing money, causing reputational damage, and performing identity theft [3], [4], [5]. Social engineering is a popular crime due to its subtle, often untraceable nature [1], [3]. Additionally, it provides the attacker with an opportunity to reap high-value rewards without using the technological tools and skills required by other cybercrimes [2].

Today there is a multitude of strong cybersecurity measures available, from firewalls and anti-virus software to cryptography and intrusion detection systems [3]. Instead of battling these technology-based protections, social engineering capitalizes on a different vulnerability: human nature [3]. Social engineers rely on the use of psychological strategies to catch the target off guard, relieve hesitation, and apply pressure to comply [2], [6]. Common tactics are urgency and trust [2], [4], [6]. Urgency involves emphasizing the gravity of the situation to provoke surprise and fear in the target. Trust can be developed by using a fake persona and authoritative language to convince the target of the attacker's legitimacy. Other strategies to strengthen social engineering attacks are subtlety, brevity, sympathy, and lingo [1], [6].

There are many different types of social engineering attacks. Some of the most common attack types are phishing, spear phishing, pretexting, help desk impersonation, BEC, and spoofing. Phishing involves sending generic social engineering messages to a large number of recipients, often via email [2], [3], [4], [5]. Spear phishing attacks are highly targeted towards a specific and often high-value individual or company [2], [3], [4], [5]. Pretexting involves using a fabrication such as a fake story or persona to deceive the target, and often coincides with other attack forms [3], [4]. Help desk impersonation is a specific type of pretext that involves instructing the victim to perform harmful actions on their electronic device under the guise of IT services [3]. BEC involves using social engineering to compromise an employee's account and then launching further attacks from that account, often while impersonating the initial victim [3], [4]. Lastly, spoofing is the creation of a fake domain that visually imitates a legitimate website or email address to trick victims into entering sensitive information [4], [5].

B. MAJOR SOCIAL ENGINEERING ATTACKS

Although social engineering attacks are very common, most schemes are minor or unsuccessful. However, it only

takes one “bite” to begin the chain of events of a high-impact operation. If an attack is successful, especially when involving a major organization, it can inflict disastrous, long-lasting, and reputation-damaging consequences.

In mid-2022, social engineering was used to obtain patients’ protected health information (PHI) from the Pennsylvania-based Allegheny Health Network [7]. A team of attackers targeted Allegheny Health Network employees using spear phishing emails. An employee subsequently clicked on the embedded malicious link and was directed to a spoofed website which harvested their login credentials. After accessing the employee’s account, the attackers downloaded roughly 8,000 patient files containing PHI including name, birth date, address, phone number, driver’s license, email address, medical conditions, diagnoses, treatment plan, and medical records. Some files also contained Social Security numbers and financial information. This data was likely sold by the attackers, which introduces the risk of identity theft for affected patients.

In early 2024, social engineering was used to steal money from the engineering and design company Arup, known for its construction of famous global architectural landmarks [8]. A group of attackers targeted Arup’s Hong Kong office using a combination of spear phishing and pretexting via deepfake technology. The attackers sent a finance employee a spear phishing email that appeared to be from Arup’s CFO. The email requested the employee’s presence in a video call regarding a secret transaction. The employee was hesitant, but the video call, which featured highly realistic deepfakes of the CFO and other familiar employees, convinced him of the transaction’s legitimacy. During the video call, the disguised attackers requested \$25.6 million USD to be sent over 15 transactions, and the employee followed through.

C. SOCIAL ENGINEERING TRENDS

In an ever-changing society, social engineering methods are pushed to evolve with the times. New attack trends form as criminals take advantage of global events, technological advances, online platforms, and vulnerable groups to devise and perpetrate stronger methods of deception.

1) LEVERAGING THE COVID-19 PANDEMIC

After the onset of the COVID-19 pandemic in March 2020, social engineers adapted their attack methods to take advantage of the new status quo. Attackers more frequently targeted healthcare organizations and hospitals with the goal of obtaining valuable pandemic-related data and sensitive patient PHI [9]. Additionally, due to the abrupt migration to online platforms, there were more opportunities for scammers to launch social engineering attacks [10]. Many employees were unfamiliar with the remote work environment and lacked online security training, which made them more vulnerable to phishing schemes [9].

Social engineers also adjusted the themes behind their attacks to take advantage of the fears and burdens caused

by the COVID-19 pandemic. Several documented phishing attacks involved impersonating government organizations such as the Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) to take advantage of people’s trust in authority [10]. Attackers also capitalized on financial crises by announcing loans, government aid, and employment updates in phishing emails to take advantage of people’s desperation [9].

2) ABUSING GENERATIVE ARTIFICIAL INTELLIGENCE

Social engineers have begun to capitalize on contemporary advancements in generative AI to strengthen the believability of their attacks. One such tool is deepfakes, or highly realistic synthetic images, videos, or audios designed to mimic real individuals [11], [12]. Social engineers have begun using deepfakes to impersonate loved ones or trusted authorities during pretexting attacks [11]. They have also started using deepfakes to create fake celebrity-endorsed promotions for social media baiting scams [11]. Another AI utility that criminals are taking advantage of is the chatbot. LLM-based chatbots such as ChatGPT are being used to generate messages with proper spelling, grammar, and tone to prevent attackers from being revealed by a language barrier [12].

3) TARGETING SOCIAL MEDIA YOUTH

Social media is a growing target for social engineers due to its wealth of information, easy sign-up process, and large youth user base. The plethora of details shared on social media is incredibly useful during the research stage of an attack [13]. In addition, profiles can divulge users’ specific interests which can be leveraged to create highly targeted attacks [13]. Furthermore, social media platforms often have a simple sign-up process, which allows social engineers to create multiple fake accounts [13]. Lastly, a majority of social media users are young adults and teenagers [13]. This age group is often inexperienced, easily convinced, and uneducated about social engineering attacks, which makes them ideal victims. Criminals often target young users by mimicking or taking over their friends’ accounts [13].

D. SOCIAL ENGINEERING STATISTICS

To track the prevalence of social engineering attacks, several organizations report annual findings based on surveys and crime reports. These papers include data regarding the general public, organizations, major breaches, counteractive measures, and notable attack strategies.

In 2023, the FBI’s Internet Crime Complaint Center (IC3) received a record number of complaints and losses, totaling 880 thousand complaints and over \$12.5 billion in losses [14]. The IC3 found that phishing was the top internet crime with over 298 thousand attacks reported, and that the costliest topics were investment and cryptocurrency scams [14]. These attacks include promises of investment coaching, lost funds recovery, and NFT promotions [14]. The IC3 found that older adults are targeted the most by internet crime, especially tech

support and government impersonation scams [14]. The 60+ age group had the highest amount of complaints and losses at over 101 thousand complaints and over \$3.4 billion in losses [14].

A 2023 study by Proofpoint found that about 71% of surveyed organizations fell victim to at least one phishing attack [12]. In a 2022 study conducted by Barracuda, spear phishing in particular affected half of the surveyed organizations, with the average organization receiving about 10 suspicious emails per day [6]. Additionally, Barracuda found that 24% of the surveyed organizations suffered an email compromise in 2022 [6]. Spear phishing was found to be a particularly dangerous threat, as it comprised about two-thirds of successful email-based social engineering attacks in the report [6].

The results of a successful attack can be devastating, especially in the event of a breach. According to a 2023 IBM report, phishing and stolen credential attacks such as BEC were the most common vectors that led to a data breach [15]. These tactics were responsible for an average of \$4.88 million in losses [15]. A 2023 report by Verizon found pretexting and phishing to be the most common vectors, contributing to 73% of breaches caused by social engineering [16].

The human element of social engineering is very important for attack resistance. The Verizon study found that 68% of breaches involved a human factor [16]. Verizon also reported that social engineering attacks can happen quickly. The survey found that the median time for an individual to fall victim to a phishing email is less than a minute: 21 seconds to click a link and 28 seconds to enter data on a malicious site [16]. Proofpoint concluded that humans are vulnerable to relaxing security practices for the sake of convenience [12]. 71% of surveyed employees responded that they had performed a risky behavior regarding cybersecurity on the job, and 96% of those employees admitted to being aware that their behavior was risky [12]. 52% of surveyed employees also responded that they are not sure if they are responsible for cybersecurity on the job [12]. These statistics highlight the need for improved security training on risks and accountability.

Based on Verizon's analysis of preventative measures, the primary shield against outside attacks is security awareness training [16]. Furthermore, IBM found that training employees to detect and stop phishing attacks was the most important factor in reducing average data breach costs [15]. However, current training methods must be improved. Proofpoint found that although 99% of surveyed employees reported to have a security awareness program at their company, only 53% reported that every employee receives training [12]. Furthermore, only 34% of surveyed employees reported that their program involves simulated phishing attacks [12]. Regarding existing simulation programs, Verizon reported that only 31% of participants were able to notice and report simulated phishing emails during awareness exercises [16]. This calls for more inclusive and effective security education, with a greater focus on simulations and in-depth learning.

Some businesses and cybersecurity professionals believe that technological solutions are sufficient to prevent successful social engineering attacks. Recent studies reveal, however, that attackers have developed methods to circumvent these measures. Barracuda found that blocklists and email authentication standards are unable to detect attacks originating from reputable domains or compromised email accounts [6]. Additionally, URL scanners can be avoided by using spoofed websites with zero-day links or compromised legitimate sites [6]. Proofpoint found that MFA bypass techniques are growing, as roughly 1 million phishing attacks using the EvilProxy MFA bypass framework are recorded each month [12]. Proofpoint also reported that a new phishing technique known as telephone-oriented attack delivery (TOAD) is on the rise [12]. TOAD attacks are messages that direct the recipient to call a phone number to continue the interaction [12]. These attacks are dangerous because, unlike URLs, attachments, and emails, phone numbers are able to avoid automated threat detection software [12]. Recently, QR codes have been leveraged by attackers for the same reason [12]. Some companies have begun using custom AI-powered attack detection software to flag anomalies by analyzing intra-organization communications [6]. However, due to the limitations of AI and the ever-evolving nature of social engineering attacks, these smart detection systems are not infallible. This underscores the significance of implementing comprehensive security awareness curriculum to defend against subtle threats.

III. EXISTING WORKS

The following section will describe similar existing works in several domains on the topic of text generation using LLMs, with discussion regarding the dataset, techniques, and best model for each work. Table 1 summarizes the findings from each work, along with best performance scores, for ease of comparison.

To the authors' abilities, no works involving social engineering scenario generation could be found. Additionally, no works involving the selected dataset source, *The Art of Deception* [17], could be found. This highlights the novelty of our study, as there is currently a knowledge gap regarding the intersection of generative modeling and cybersecurity education. Although there is significant existing exploration into defensive methods, such as AI-powered attack detection, there is a lack of literature that involves offensive attack modeling and generation.

In a 2020 study by Kapočiūtė-Dzikiene [18], the author developed a domain-specific chatbot using a small training dataset translated into both English and Lithuanian. The dataset consisted of 567 question-answer pairs about a software company's products, prices, technologies, and other relevant information. The dataset was preprocessed using punctuation removal and punctuation separation to isolate words. Word embeddings were obtained using combinations of three different methods: One-hot encoding, FastText, and BERT. Three different seq2seq model architectures were

TABLE 1. Summary of Existing Research on Text and Scenario Generation Using LLMs Across Application Domains.

Paper	Dataset(s)	Techniques	Best Model(s)	Best Performance Scores
Kapočiūtė-Dzikiene 2020 [18]	Company info QA pairs in English and Lithuanian	Seq2seq frameworks LSTM BiLSTM Word embeddings One-hot encoding FastText BERT	One-layer BiLSTM encoder One-layer LSTM decoder BERT embeddings	BLEU: 0.513 ROUGE-2 F1: 0.495 ROUGE-L F1: 0.565
Qu <i>et al.</i> 2020 [19]	Chinese articles Chinese blog posts	Char-level splitting BERT tokenization	Text Generation: GPT-2 Text Prediction: ChineseBERT	Human Evaluation: Appropriate topic relevance Appropriate readability
Li <i>et al.</i> 2021 [20]	Articles Documents Question-answer pairs Forums Dialogue Stories	Source prompt pooling Prompt clustering Multi-key memory network Adaptive attention mechanism	Prompt transfer for Text Generation (PTG)	ROUGE-1: 42.40 ROUGE-2: 21.35 ROUGE-L: 39.14 BLEU-1: 45.46 BLEU-2: 29.52 Distinct-1: 1.46 Distinct-2: 8.34
Perdana <i>et al.</i> 2022 [21]	Brawijaya University FAQ Human-annotated and T5-generated QA pairs	Rasa NLU Intent classification Context tracking	Intent Evaluation: Rasa NLU + IndoBERT Response Retrieval: Rasa NLU	Intent Evaluation: Accuracy: 0.961 F1: 0.958 Precision: 0.965 Response Retrieval: Accuracy: 0.835 F1: 0.824 Precision: 0.863
Anaya <i>et al.</i> 2024 [22]	Biometrics QA pairs Gene ontology Pharmaceutical info Biometrics forum QA pairs	QLoRA Parameter-efficient fine-tuning Prompt learning	Mistral-7B-Instruct-v0.1	Yes/No Questions: Accuracy: 0.4074 F1 Yes: 0.5294 F1 No: 0.2 Macro F1: 0.3647 Factoid Questions: Strict Accuracy: 0.1579 Lenient Accuracy: 0.1579 Mean Reciprocal Rank: 0.1579 Summary Questions: ROUGE-2 Recall: 0.0845 ROUGE-2 F1: 0.0543 ROUGE-SU4 Recall: 0.1172 ROUGE-SU4 F1: 0.0719
Chang <i>et al.</i> 2024 [23]	Risky highway scenarios	Scenario prompting Scenario generation	GPT-4	Rarity: 23 Reality: 24
Park <i>et al.</i> 2024 [24]	Few-shot query learning	Query Transformation Module General Query Preceding Phrases Query Cloze Query Purpose Explicit Query	Korean GPT-3 Purpose Explicit Query	Sensibleness: 0.578 Specificity: 0.489

tested: a one-layer LSTM encoder with a one-layer LSTM decoder, a two-layer stacked LSTM encoder with a two-layer LSTM decoder, and a one-layer BiLSTM encoder with a one-layer LSTM decoder. The architecture with the best performance using the English version of the dataset was the third architecture coupled with BERT word embeddings for both halves.

In a 2020 report by Qu *et al.* [19], the authors developed a generic text generation and prediction system. The dataset consisted of articles and blog posts written in Chinese, and was comprised of text from the BaiduBaike and LLKT websites. The data was preprocessed using char-level splitting to isolate Chinese characters and BERT tokenization to transform the text sequences into vectors. For the text generation model, GPT-2 was used to produce new text based on simple starting words, and for the text prediction

model, ChineseBERT was used to make word predictions based on surrounding words. Based on human evaluation, the topic relevance and readability were considered appropriate for both models. Notable observations were that the LLKT dataset produced stronger models and that the GPT-2 model performed worse when generating downward while being fed previously generated text.

In a 2021 paper by Li *et al.* [20], the authors developed a new prompt-based method for transferable text generation using pre-trained LLMs. The dataset consisted of articles, documents, question-answer pairs, forums, dialogue, and stories. These texts were divided into three generation tasks: compression (summarization and question generation), transduction (style transfer and paraphrasing), and creation (conversational responses and story generation). The authors used source prompt pooling via prompt clustering to store

reusable prompts for each task. These prompts were then stored in a multi-key memory network with an adaptive attention mechanism to form a novel Prompt Transfer for Text Generation (PTG) model for flexible transfer learning. PTG utilizes three LLMs, one for prompt learning (BART) and two for text generation (BART and BERT). This architecture was compared to six different models: GPT-2, BART, T5, Prefix-Tuning, SPoT, and multi-task fine-tuned BART. The model with the best overall performance was PTG.

In a 2022 study by Perdana et al. [21], the authors developed a domain-specific chatbot using a low-resource language. The dataset consisted of FAQ about Brawijaya University written in Indonesian: 580 human-annotated question-answer pairs and 1160 T5-generated question-answer pairs. The authors incorporated Rasa NLU into their models for intent classification and conversation context tracking. Four different architectures were created and tested for intent evaluation and response retrieval: Rasa NLU alone, Rasa NLU with IndoBERT phase 1, Rasa NLU with IndoBERT phase 2, and Rasa NLU with Multilingual BERT. The third architecture yielded the best performance for intent evaluation and the first architecture yielded the best performance for response retrieval.

In a 2024 report by Anaya et al. [22], the authors developed a domain-specific chatbot using open-source data. Four datasets were used: BioASQ (biometrics question-answer pairs), Gene Ontology (biometrics and gene-related definitions), DrugBank (pharmaceutical information), and BiQA (biometrics forum questions and cited article answers). The authors used the Mistral-7B-Instruct v0.1 model with QLoRA for parameter efficient fine-tuning and prompt learning to optimize performance. After the model was fine-tuned, its performance was evaluated using three types of questions: yes/no, factoid, and summary.

In a 2024 paper by Chang et al. [23], the authors used LLMs to develop a scenario generation chatbot. The dataset consisted of real-life risky highway driving scenarios, and the authors' goal was to generate unique risky scenarios as corner cases for autonomous vehicle testing. The scenario prompting stage of development involved using instruction tokens to describe the LLM's role, scenario tokens to describe example scenarios, and experience tokens to help the LLM improve based on past generated scenarios. The scenario generation stage of development involved solidifying the LLM's role and task using chain-of-thought questioning, requiring the LLM's output to be in trajectory format, instructing the LLM to include descriptions and risk analysis, and tuning the LLM through evaluation and feedback. Three different models were tested: GPT-4, OpenChat-3.5-7b, and Llama-2-7b, and the model with the best performance was GPT-4.

In a 2024 study by Park et al. [24], the authors developed a conversational generative AI model using prompt engineering with LLMs. The authors did not use a traditional dataset, as they were focused on prompt design with few-shot learning. The authors used a query transformation module (QTM) with four query types: general query (no

adjustments), preceding phrases query (initial context words), cloze query (fill-in-the-blank with examples), and purpose explicit query (objective with examples). The selected LLMs were the Korean versions of GPT-2 (SKT GPT-2) and GPT-3 (Kakaobrain KoGPT-3). The model with the best performance was Korean GPT-3 with the purpose explicit query technique.

IV. TECHNOLOGIES

The following section will describe the technologies used for the project, namely language and environment.

A. LANGUAGE

The language used for the following experiments is Python 3. Python was primarily selected for its various libraries, modules, and frameworks that support data manipulation, data visualization, and machine learning. Additionally, the API documentation for GPT-3.5 and Llama 3.1 feature code snippets written in Python. Other benefits of Python include its straightforward syntax and open source nature.

B. ENVIRONMENT

The environment used for the following experiments is Google Colab Pro [25]. Google Colab offers hosted Jupyter Notebook services for users with a Google account [25]. The Google Colab environment is advantageous due to its simple setup, cloud-based resources, and Google Drive connectivity [25]. Colab Pro is a paid service tier that provides access to more powerful hardware accelerators using the Google Compute Engine backend [25]. For these experiments, the A100 GPU was used, which allocates 83.5 GB of system RAM, 40 GB of GPU RAM, and 112.6 GB of disk space [25].

V. DATASET

There are few options for public data involving social engineering attacks. As discussed in section II-D, the FBI and other cybersecurity organizations collect records of social engineering attacks, yet aside from generalized annual reports, the data is strictly confidential for the privacy of citizens and companies. On the internet, there are various articles, videos, and screenshots that detail social engineering attacks, yet this data is raw, sparse, and varies in quality. Instead of undertaking the feat of processing and compiling a dataset from scratch, the authors chose to utilize a famous book containing social engineering scenarios.

The selected dataset is a small collection of 45 social engineering scenarios from the book *The Art of Deception* [17]. These scenarios involve an attacker using various social engineering techniques to successfully manipulate the target and achieve their goal. The scenarios are written in English and most contain a significant amount of conversational dialogue. Although the scenarios are stated to be fictional in nature, they are heavily inspired by real attacks encountered by the author, who is a former social engineer and hacker.



FIGURE 1. Distributions of Various Social Engineering Communication Modes, Types, Strategies, and Goals across the Dataset Scenarios.

We analyzed these scenarios to determine the various social engineering techniques that they feature. Within the dataset, the following modes of communication are present: phone call, in person, and email [17]. The dataset also features the use of the following social engineering types: pretexting, spear phishing, help desk impersonation, physical breach attack, website spoofing, phishing, baiting, caller ID spoofing, and shoulder surfing [17]. Within the dataset, the following strategies are present: sympathy, insider knowledge, lingo, urgency, flattery, malware (Trojan horse and spyware), password attack (dictionary and brute force), forgery, and breaking and entering [17]. The dataset scenarios feature goals which concern sensitive information, system access, identity theft, defamation, trespassing, monetary gain, material theft, and personal gain [17]. The number of scenarios that involve each class of communication mode, type, strategy, and goal are visualized using bar plots in Fig. 1. Based on these distributions, the most common mode of communication is phone call (present in 40 scenarios), the most common attack types are pretexting and spear phishing (present in 41 and 40 scenarios, respectively), the most common strategies are sympathy, insider knowledge, and lingo (present in 15, 15, and 14 scenarios, respectively), and the most common goal is obtaining sensitive information (present in 28 scenarios). Due to the complexity of the dataset, many scenarios fall into more than one class in each category, that is, a given scenario may involve multiple different modes, types, strategies, or goals. Therefore, due to this overlap, the scenario counts for each class within a graph do not add up to the total number of scenarios in the dataset, which is 45.

VI. DATA PREPROCESSING

The following section will describe the preprocessing steps used to prepare the scenarios dataset for integration with an LLM.

A. EXTRACTION

The scenarios were obtained by reading *The Art of Deception* [17] and manually extracting scenarios from the text. The excerpts were copied from the book and transferred to a word processing document for further editing.

B. CLEANING

To clean the dataset, we formatted the scenarios uniformly. We introduced section headings for different interactions and story scenes for improved organization. We also added speaker labels for dialogue clarity, as not all speech was labeled by the book authors. We altered the text so that any reference to the attacker was indicated using the word “attacker” instead of the aliases found in the book. We also removed all irrelevant scenario content, which includes background information about the involved parties, extraneous storyline details, and author commentary about the attack. Many scenarios contain background information about the victim (their job, their personal characteristics, how their day is going at the time of the attack) and the attacker (their motivation, their preparation for the attack, and their personal life). All of this information was removed from the scenario text, as it is irrelevant to the active execution of the social engineering attack. Many scenarios also contain extraneous storyline language that describes redundant or

insignificant details, such as typing out a code or hanging up the phone. This content was removed, as although these details are helpful for painting a realistic story, they do not contribute to story progression. Lastly, most scenarios contain author commentary, namely “Mitnick Message,” which gives an overarching conclusion about the nature of the attack, “Analyzing the Con,” which dissects the attack from the perspectives of the attacker and victim(s), and “Preventing the Con,” which discusses how to prevent falling victim to such an attack. Despite the valuable insight provided by this analysis, it was removed, as it is not applicable to the events of the attack itself.

C. INSTRUCTIONS

One-sentence instruction prompts were designed to accompany the dataset scenarios during the fine-tuning process for the BERT, T5, and GPT-3.5 models. Each instruction was written to be intentionally simple and direct, while adhering to the format: “Create a scenario where an attacker [malicious action].” During fine-tuning, these instructions and their corresponding scenarios were passed to the models as a dataset of inputs and outputs, to teach the models to generate in-depth scenarios based on a short description of the attacker’s goal. The role of these instructions was to supplement the dataset rather than directly guide prompt-based generation, so no extensive prompt validation was incorporated.

D. FORMATTING

The scenarios were copied from the word processing document and stored in CSV format. Two file versions were maintained, one CSV contained instructions and scenarios and the other contained only scenarios. For the BERT and T5 models, the dataset was loaded from the CSV file containing instructions and scenarios. For compatibility with the GPT-3.5 model, the dataset (with instructions) was converted from CSV to JSONL format. For compatibility with the Llama 3.1 model, the dataset (without instructions) was converted from CSV to JSON format.

E. SAMPLE SCENARIO

Table 2 includes a sample scenario from the dataset. It includes both the raw scenario text from *The Art of Deception* [17] as well as the corresponding preprocessed version. This provides context to the style, format, and content of the dataset scenarios, as well as insight into the changes performed during preprocessing. Note that the author commentary associated with this scenario was omitted in favor of length. Due to copyright, requests to access the entire dataset should be directed to Kevin D. Mitnick and William L. Simon, the authors of *The Art of Deception* [17].

VII. EXPERIMENTS

The following section will describe the experiments involving the BERT, T5, GPT-3.5, and Llama 3.1 models.

A. EXPERIMENT PIPELINE

The experiment pipeline is shown in Fig. 2. For each experiment, the preprocessed and formatted dataset is fed to a pre-trained LLM. For the BERT, T5, and GPT-3.5 models, fine-tuning is performed using the training and validation sets to guide the model for the task of scenario generation. For the Llama 3.1 model, prompt engineering is performed to guide the model for the task of scenario generation. Each model’s training performance is evaluated using either metrics and loss minimization or only metrics. If the performance is suboptimal, the model hyperparameters (and instructions if applicable) are adjusted, and the training process is repeated. Once an optimal performance is reached, the best configuration for each model is saved, and each model’s testing performance is evaluated using various metrics.

B. BERT

1) DESCRIPTION

Bidirectional Encoder Representations from Transformers, known as BERT, is an LLM developed by Google in 2018 [26]. It is a transformer-based, encoder-only deep learning model that was pre-trained using masked language modeling and next sentence prediction to support contextual understanding [26]. The selected model version for the BERT experiment is BERT Base Uncased, which has 110 million parameters and a context window of 512 tokens [27]. BERT Base Uncased was pre-trained on 3.3 billion tokens from the English Wikipedia and BookCorpus [28] datasets [26].

2) MODEL-SPECIFIC PREPROCESSING

The scenarios and instructions were loaded from the CSV dataset file and converted to a dataset dictionary. Shuffled hold-out was performed with 90% of data reserved for training and 10% of data reserved for validation. Due to the small size of the dataset, the hold-out ratio favors the training dataset in order to devote a strong majority of samples for direct model tuning. A test set was not reserved because the testing prompts were written from scratch. The BERT Base tokenizer was used to tokenize the dataset and obtain the inputs (instructions), labels (scenarios), and attention mask.

3) FINE-TUNING

Hugging Face’s Transformers library was used to perform supervised fine-tuning on the BERT Base model to guide it to receive an instruction and generate a scenario that matches the prompt [29]. The fine-tuning process was repeated in order to perform hyperparameter tuning with the goal of improving model performance. Model performance was analyzed based on loss minimization and output quality. Output quality was determined via the metrics described in Section VIII-A. The following hyperparameters were tuned: max length, min length, no repeat N-gram size, num beams, length penalty, early stopping, temperature, optimizer, learning rate, patience, and batch size [30]. Max and min length were used to prevent unusually long or short

TABLE 2. Sample Dataset Scenario in Raw and Preprocessed Forms.

Raw Scenario from <i>The Art of Deception</i> [17]	Scenario After Preprocessing
Janie Acton's Story Janie Acton had been manning a cubicle as a customer service rep of Hometown Electric Power, in Washington, D.C., for just over three years. She was considered to be one of the better clerks, smart and conscientious. It was Thanksgiving week when this one particular call came in. The caller, said, "This is Eduardo in the Billing Department. I've got a lady on hold, she's a secretary in the executive offices that works for one of the vice presidents, and she's asking for some information and I can't use my computer. I got an email from this girl in Human Resources that said 'ILOVEYOU,' and when I opened the attachment, I couldn't use my machine any more. A virus. I got caught by a stupid virus. Anyways, could you look up some customer information for me?" "Sure," Janie answered. "It crashed your computer? That's terrible." "Yeah." "How can I help?" Janie asked. Here the attacker called on information from his advance research to make himself sound authentic. He had learned that the information he, wanted was stored in something called the Customer Billing Information System, and he had found out how employees referred to the system. He asked, "Can you bring up an account on CBIS?" "Yes, what's the account number?" "I don't have the number; I need you to bring it up by name." "Okay, what's the name?" "It's Heather Marning." He spelled the name, and Janie typed it in. "Okay, I have it up." "Great. Is the account current?" "Uh huh, it's current." "What's the account number?" he asked. "Do you have a pencil?" "Ready to write." "Account number BAZ6573NR27Q." He read the number back and then said, "And what's the service address?" She gave him the address. "And what's the phone?" Janie obligingly read off that information, too. The caller thanked her, said good-bye, and hung up. Janie went on to the next call, never thinking further about it.	Janie's Caller Attacker: "This is Eduardo in the Billing Department. I've got a lady on hold, she's a secretary in the executive offices that works for one of the vice presidents, and she's asking for some information and I can't use my computer. I got an email from this girl in Human Resources that said 'ILOVEYOU,' and when I opened the attachment, I couldn't use my machine any more. A virus. I got caught by a stupid virus. Anyways, could you look up some customer information for me?" Janie: "Sure. It crashed your computer? That's terrible." Attacker: "Yeah." Janie: "How can I help?" Attacker: "Can you bring up an account on CBIS?" Janie: "Yes, what's the account number?" Attacker: "I don't have the number; I need you to bring it up by name." Janie: "Okay, what's the name?" Attacker: "It's Heather Marning." Janie: "Okay, I have it up." Attacker: "Great. Is the account current?" Janie: "Uh huh, it's current." Attacker: "What's the account number?" Janie: "Do you have a pencil?" Attacker: "Ready to write." Janie: "Account number BAZ6573NR27Q." The attacker read the number back and then said "And what's the service address?" Janie gave him the address. Attacker: "And what's the phone?" Janie gave him the phone number too.

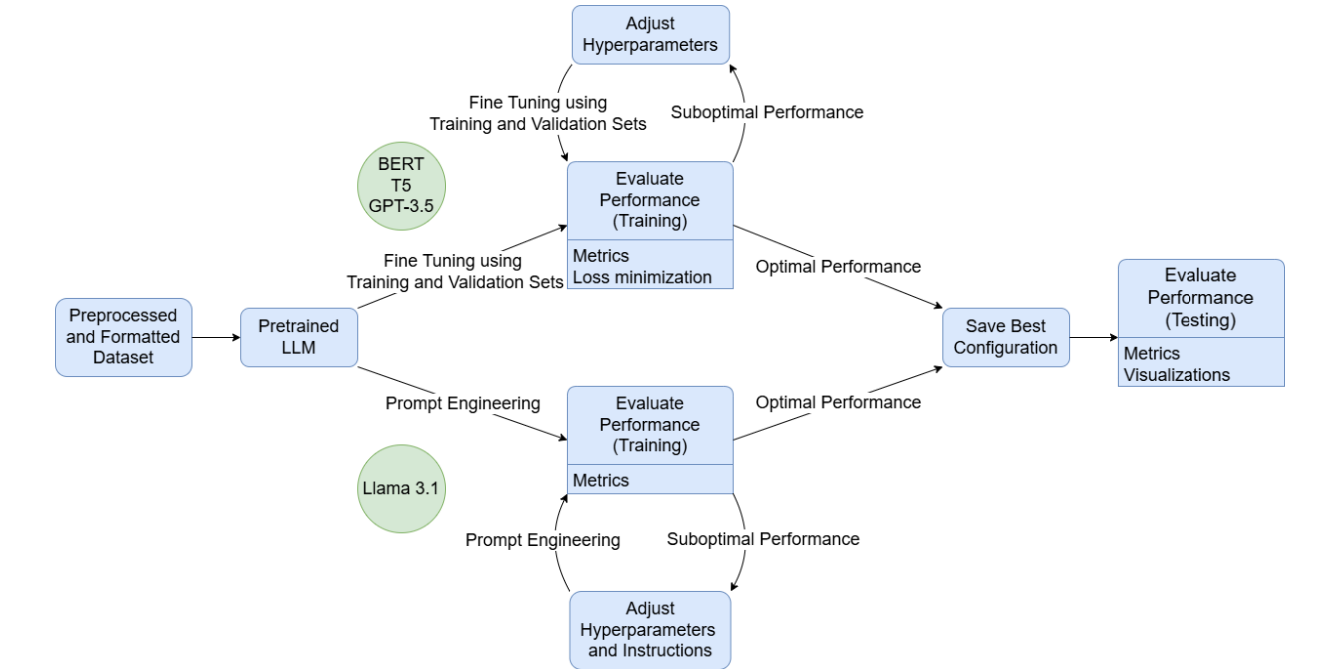


FIGURE 2. Experiment Pipeline.

TABLE 3. Optimal BERT Hyperparameters.

Hyperparameter	Value
Max Length	512
Min Length	64
No Repeat N-Gram Size	5
Num Beams	4
Length Penalty	2
Early Stopping	True
Temperature	1
Optimizer	AdamW
Learning Rate	0.0001
Patience	5
Batch Size	5

outputs by controlling the number of tokens generated by the model. No repeat N-gram size was used to encourage varied language by preventing multiple occurrences of 5-grams in the generated outputs. Num beams, length penalty, and early stopping relate to the beam search generation technique, which was utilized for its efficient maintenance of several candidate sequences. Temperature was used to encourage unique and non-deterministic outputs. The chosen optimizer was AdamW with a learning rate of 0.0001, to encourage better generalization and more precision. Patience was used to prevent overfitting. The optimal hyperparameter values are shown in Table 3.

C. T5

1) DESCRIPTION

Text-To-Text Transfer Transformer, known as T5, is an LLM developed by Google in 2019 [31]. It is a transformer-based, encoder-decoder deep learning model that utilizes a unified framework to process all NLP tasks using a text-to-text format [31]. The selected model version for the T5 experiment is T5 Base, which has 223 million parameters and a context window of 512 tokens [32]. T5 Base was pre-trained on the Colossal Clean Crawled Corpus (C4) which contains 156 billion tokens of clean web-scraped English text [32].

2) MODEL-SPECIFIC PREPROCESSING

The scenarios and instructions were loaded from the CSV dataset file and converted to a dataset dictionary. Shuffled hold-out was performed with 90% of data reserved for training and 10% of data reserved for validation. As with the BERT experiment, the hold-out ratio favors the training dataset in order to devote a strong majority of samples for direct model tuning. A test set was not reserved because the testing prompts were written from scratch. The T5 Base tokenizer was used to tokenize the dataset and obtain the inputs (instructions), labels (scenarios), and attention mask.

3) FINE-TUNING

As seen in the BERT experiment, the Transformers library by Hugging Face was used to perform supervised fine-tuning on the T5 Base model to guide it to receive an instruction

TABLE 4. Optimal T5 Hyperparameters.

Hyperparameter	Value
Max Length	1024
Min Length	64
No Repeat N-Gram Size	5
Num Beams	5
Length Penalty	2
Early Stopping	True
Temperature	1
Optimizer	AdamW
Learning Rate	0.0001
Patience	5
Batch Size	5

and generate a scenario that matches the prompt [29]. The fine-tuning process was repeated in order to perform hyperparameter tuning with the goal of improving model performance. Model performance was analyzed based on loss minimization and output quality. Output quality was determined via the metrics described in Section VIII-A. The following hyperparameters were tuned: max length, min length, no repeat N-gram size, num beams, length penalty, early stopping, temperature, optimizer, learning rate, patience, and batch size [30]. These hyperparameters were used in the same manner as seen with the BERT model, which is described in Section VII-B3. The optimal hyperparameter values are shown in Table 4.

D. GPT-3.5

1) DESCRIPTION

Generative Pre-trained Transformer 3.5, known as GPT-3.5, is an LLM developed by OpenAI in 2022 [33]. It is a transformer-based, decoder-only deep learning model that was pre-trained using reinforcement learning from human feedback (RLHF) to support more user-friendly and conversational outputs [33], [34]. The selected model version for the GPT-3.5 experiment is GPT-3.5 Turbo 0125, which is a version that is fast, is tailored for conversational dialogue, and can be fine-tuned on a dataset [35]. GPT-3.5 Turbo 0125 has a context window of 16k tokens and a knowledge cutoff of September 2021 [33]. The number of parameters and the size of the pre-training dataset for GPT-3.5 are unknown, but for reference, GPT-3 has 175 million parameters and was pre-trained on 500 billion tokens from the Common Crawl [36], WebText2 [37], Books1, Books2, and Wikipedia datasets [38].

2) MODEL-SPECIFIC PREPROCESSING

The scenarios and instructions were loaded from the JSONL training and validation dataset files, with a ratio of 90% for training and 10% for validation. As with the previous experiments, the hold-out ratio favors the training dataset in order to devote a strong majority of samples for direct model tuning. A test set was not reserved because the testing prompts were written from scratch. Tokenization was not necessary.

TABLE 5. Optimal GPT-3.5 Hyperparameters.

Hyperparameter	Value
Max Tokens	2000
N	1
Frequency Penalty	0
Presence Penalty	0
Top P	0.5
Temperature	1
Learning Rate Multiplier	2
Batch Size	5

3) FINE-TUNING

An OpenAI client was used to perform supervised fine-tuning on the GPT-3.5 Turbo model to guide it to receive an instruction and generate a scenario that matches the prompt [39]. The fine-tuning process was repeated in order to perform hyperparameter tuning with the goal of improving model performance. Model performance was analyzed based on loss minimization and output quality. Output quality was determined via the metrics described in Section VIII-A. The following hyperparameters were tuned: max tokens, N, frequency penalty, presence penalty, top P, temperature, learning rate multiplier, and batch size [40]. Max tokens was used to prevent especially long outputs by limiting the number of tokens generated by the model. N was used to specify the number of outputs to generate. The frequency and presence penalties were used to affect the generation of diverse language by penalizing or non-penalizing new tokens based on their current appearance in the output. Top P and temperature were used to encourage unique, varied, and non-deterministic outputs. The optimal hyperparameter values are shown in Table 5.

E. LLAMA 3.1

1) DESCRIPTION

Large Language Model Meta AI 3.1, known as Llama 3.1, is an LLM developed by Meta in 2024 [41]. It is a transformer-based, decoder-only deep learning model that utilizes a grouped-query attention mechanism for improved inference capabilities and supports a wide range of generation tasks [42]. The selected model version for the Llama 3.1 experiment is Llama 3.1 8B Instruct, which has 8 billion parameters, a context window of 128k tokens, and a knowledge cutoff of December 2023 [43]. Llama 3.1 8B Instruct was pre-trained using over 15 trillion tokens of unspecified public data gathered from the web [43].

2) MODEL-SPECIFIC PREPROCESSING

The scenarios were loaded from the JSON dataset file and stored as a dictionary. Hold-out and tokenization were not necessary.

3) PROMPT-ENGINEERING

Instead of fine-tuning, the Llama 3.1 experiment involved the technique of prompt engineering using LangChain's ChatPromptTemplate, which supports the development of flexible

TABLE 6. Optimal Llama 3.1 Hyperparameters.

Hyperparameter	Value
Max Tokens	2000
Frequency Penalty	1
Presence Penalty	0
Top P	0.2
Temperature	0.5

and specialized prompt templates with roles, instructions, input placeholders, and chat history [44]. Two model versions were created, a one-step scenario generator and an interactive scenario generator. For both model versions, the prompt engineering process involved tuning the prompt instructions and the model hyperparameters with the goal of improving model performance. Model performance was analyzed based on output quality, which was determined via the metrics described in Section VIII-A. The following hyperparameters were tuned: max tokens, frequency penalty, presence penalty, top P, and temperature [45]. These hyperparameters were used in the same manner as seen with the GPT-3.5 model, which is described in Section VII-D3. The optimal hyperparameter values are shown in Table 6.

a: ONE-STEP SCENARIO GENERATION

For the one-step scenario generator, the prompt instructs Llama 3.1 to create a unique and complete scenario that is similar in style and format to the scenarios found in the dataset. The prompt also asserts that the scenario must involve an attacker that leverages social engineering techniques found in the dataset. The prompt maintains that this attacker must successfully achieve the user-specified goal by the end of the scenario. Despite this, the prompt asserts that the victim must not be easily deceived (such as by a blatant or simple attack), yet they are still susceptible to be inevitably manipulated through social engineering.

b: INTERACTIVE SCENARIO GENERATION

For the interactive scenario generator, the prompt instructs Llama 3.1 to build a unique scenario similar to the scenarios found in the dataset. Instead of generating a complete scenario in one step, the prompt directs Llama 3.1 to ask the user one question at a time to obtain details as it gradually develops a custom-tailored scenario. Chain-of-thought prompting is utilized, as Llama 3.1 is instructed to explain each step of the scenario as it is produced, as well as ask the user for feedback. The prompt asserts that the first question asked by Llama 3.1 should always involve the attacker's goal, in order to set the stage for the scenario. Furthermore, the prompt instructs Llama 3.1 to guide the scenario in a direction that results in the attacker's success regardless of potential user-specified obstacles. This implies that the attacker should be highly skilled in various social engineering tactics. In this regard, the prompt maintains that Llama 3.1 must incorporate social engineering techniques found in the dataset and that the victim must not be easily

fooled. Furthermore, the prompt dictates that if the user is ever unsure of how to proceed with the scenario, Llama 3.1 should propose a way to move the story forward. Once the attacker has achieved their goal, the prompt instructs Llama 3.1 to end the scenario and rewrite it from start to finish, in a style and format similar to the dataset samples. The prompt features a record of chat history to enable this interactive, step-by-step conversation between the user and Llama 3.1.

VIII. RESULTS

The following section will describe the results of the experiments involving the tuning of BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive for the task of scenario generation.

A. METRICS

A range of metrics were used to judge the performance of the five LLM configurations defined in this study. Nine style metrics were used to analyze the characteristics of the model outputs, and four evaluation metrics were used to assess the quality of the model outputs as scenarios. Every metric used in this study is a well-known benchmark for assessing generated text to determine model performance. Our reason for incorporating such a breadth of metrics is to create robust assessments that highlight the use cases of multiple recognized evaluation techniques.

1) STYLE METRICS

The style metrics used in this study are character count, word count, sentence count, type-token ratio (1), Flesch Reading Ease (2), polarity (3), cosine similarity (4) (5), first order coherence (6), and second order coherence (7). These metrics were selected for their recognition as NLP standards for measuring text attributes. Character count, word count, and sentence count indicate the length of the text. Type-token ratio represents lexical diversity, where a higher ratio indicates more word variety. Flesch Reading Ease represents readability, where a score of 70 or higher indicates easy readability for the average adult. Polarity, which is calculated using VADER Sentiment's compound valence scoring [46], represents sentiment. Cosine similarity represents average semantic similarity to the dataset, where a higher score indicates stronger similarity. In order to calculate cosine similarity, Term Frequency-Inverse Document Frequency (TF-IDF) was used for text vectorization [47]. Coherence represents logical flow between consecutive sentences (first order) or sentences two hops apart (second order), and a higher score indicates stronger sentence relatedness.

$$TTR = \text{types}/\text{tokens} \quad (1)$$

types: number of distinct words in the text
tokens: number of words in the text

range: (0, 1]

$$FRE = 206.835 - 1.015\left(\frac{w}{t}\right) - 84.6\left(\frac{s}{w}\right) \quad (2)$$

w: number of words in the text

t: number of sentences in the text

s: number of syllables in the text

range: (0, 100]

positive polarity: $c \geq 0.5$

negative polarity: $c \leq -0.5$

neutral polarity: $c > -0.5$ & $c < 0.5$ (3)

c: compound score

range: [-1, 1]

$$\text{averageCosineSimilarity} = \frac{\sum_{i=1}^n c_i}{n} \quad (4)$$

c_i: cosine similarity between the target and a corpus text

n: number of texts in the corpus

range: [0, 1]

$$\text{cosineSimilarity} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \times \|\vec{y}\|} \quad (5)$$

$\vec{x}$: vectorized text #1

$\vec{y}$: vectorized text #2

range: [0, 1]

$$\text{firstOrderCoherence} = \frac{\sum_{i=1}^n c_i}{n} \quad (6)$$

c_i: cosine similarity for a pair of consecutive sentences

n: number of pairs of consecutive sentences

range: [0, 1]

$$\text{secondOrderCoherence} = \frac{\sum_{i=1}^n c_i}{n} \quad (7)$$

c_i: cosine similarity for a pair of sentences two hops apart

n: number of pairs of sentences two hops apart

range: [0, 1]

2) EVALUATION METRICS

Four evaluation metrics were used in this study; three metrics involve human evaluation, and the fourth metric is LLM-as-a-judge.

Human evaluation was used to assess whether the generated outputs align with the high-level requirements of a good social engineering scenario. We determined that such a scenario must follow the prompt, result in the attacker's success, and involve recognizable social engineering techniques. Therefore, we defined the human evaluation metrics to be prompt adherence, attacker success, and techniques used. Prompt adherence is a binary metric (True/False) that indicates whether the scenario follows its prompt. Attacker success is a binary metric (True/False) that indicates whether the attacker in the scenario achieves the specified goal. Techniques used is a qualitative metric that encompasses all of the social engineering types and strategies used in the scenario. All human evaluations were performed by the

same individual, J. Webb, for consistency. We decided to incorporate human evaluation because it is a standard method for analyzing the overarching, non-formula-based qualities of generated text.

LLM-as-a-judge is a contemporary method for evaluating LLM-based conversational assistants. This metric involves using an LLM to examine and score artificially generated text [48]. It is a useful technique for assessing open-ended outputs that lack a concrete target for comparison, such as the scenarios in this study [48]. In addition, this approach provides automated and detailed evaluations that align with human preferences without the costs of manual human assessment [48].

For our judge, we chose the Mixtral-8 × 7b Instruct model, which has 46.7 billion parameters, a context window of 32k tokens, and an undisclosed knowledge cutoff [49]. We selected this model to be our judge, because it is optimized for instruction following with reduced bias, which is well suited for conducting fair judgments that align with the guidelines provided in the prompt [50]. Mixtral-8 × 7b Instruct also possesses advanced reasoning and creativity, which are useful qualities for developing thorough and rational explanations for its evaluations [51]. Furthermore, Mixtral-8 × 7b Instruct uses a sparse mixture of experts to maintain efficiency despite its large parameter count [50]. This architecture is ideal for a judge, as it is less resource intensive while still achieving competitive performance [50]. The tasks required by our judge are also not demanding enough to require the use of a top-ranking model like GPT-4o.

LLM-as-a-judge was implemented using prompt engineering with LangChain, in the same manner as the Llama 3.1 experiment. The prompt instructs Mixtral-8 × 7b to analyze a prompt and a scenario, then produce a score to represent the quality of the scenario, and provide additional comments as justification. The scoring scale is described below.

- 1: Incomprehensible, irrelevant
- 2: Missing major prompt elements, different format
- 3: Short and simple, addresses aspects of the prompt
- 4: Complete, follows the prompt, format matches the dataset
- 5: Detailed, well-crafted, excellent use of social engineering

B. DATASET ANALYSIS

To establish a characteristic baseline, the dataset was evaluated using the following style metrics: character count, word count, sentence count, type-token ratio, Flesch Reading Ease, polarity, first order coherence, and second order coherence. The score distributions are visualized using histograms in Fig. 3. Based on the histograms, most dataset scenarios are fairly short, have a medium level of word variety, are easy for the average adult reader, have a strongly negative sentiment, and have a fairly high level of logical flow.

TABLE 7. Test Set of Five Scenario Generation Prompts.

Sample	Instruction
1	Create a scenario where an attacker steals an email username and password.
2	Create a scenario where an attacker enters a locked facility.
3	Create a scenario where an attacker obtains access to the company database and edits important data.
4	Create a scenario where an attacker downloads sensitive source code files.
5	Create a scenario where an attacker steals money from the victim through email.

TABLE 8. BERT Test Scores - Style Metrics.

Output	Character Count	Word Count	Sentence Count	Type-Token Ratio	Flesch Reading Ease
1	2053	412	35	0.24	94.15
2	1847	423	53	0.23	98.01
3	2044	415	21	0.24	68.5
4	2078	400	37	0.21	70.39
5	2247	424	36	0.2	77.33

TABLE 9. BERT Test Scores - Style Metrics (Cont.).

Output	Polarity	Cosine Similarity	First Order Coherence	Second Order Coherence
1	0.89	0.04	0.78	0.83
2	-0.86	0.1	0.8	0.8
3	-1	0.18	0.85	0.82
4	-1	0.3	0.77	0.76
5	-0.99	0.28	0.81	0.81

C. TESTING DATA

The test set consists of five one-sentence instructions to serve as generation prompts for each LLM configuration, as shown in Table 7. For the Llama 3.1: One-Step experiment, the shortened instructions were passed as input to the existing prompt. For the Llama 3.1: Interactive experiment, the instructions were inputted by the test user early on in the conversation.

D. BERT EVALUATION

The style metric scores for the five scenarios generated by BERT are shown in Tables 8 and 9. Note that the median for each metric is bolded. When compared to the baseline dataset, the scenarios generated by BERT are generally longer, have a much lower level of lexical diversity, are slightly easier to read, have a similarly negative sentiment, are strongly semantically dissimilar, and have a similarly high degree of logical flow. Overall, these scores are desirable except for the high sentence count, low lexical diversity, and low cosine similarity. The scores for these style metrics suggest that BERT created overly long scenarios that lack unique content and vary excessively from the dataset.

The human evaluation metrics for the five scenarios generated by BERT are shown in Table 10. These evaluations are very poor, as BERT failed to generate any scenarios that

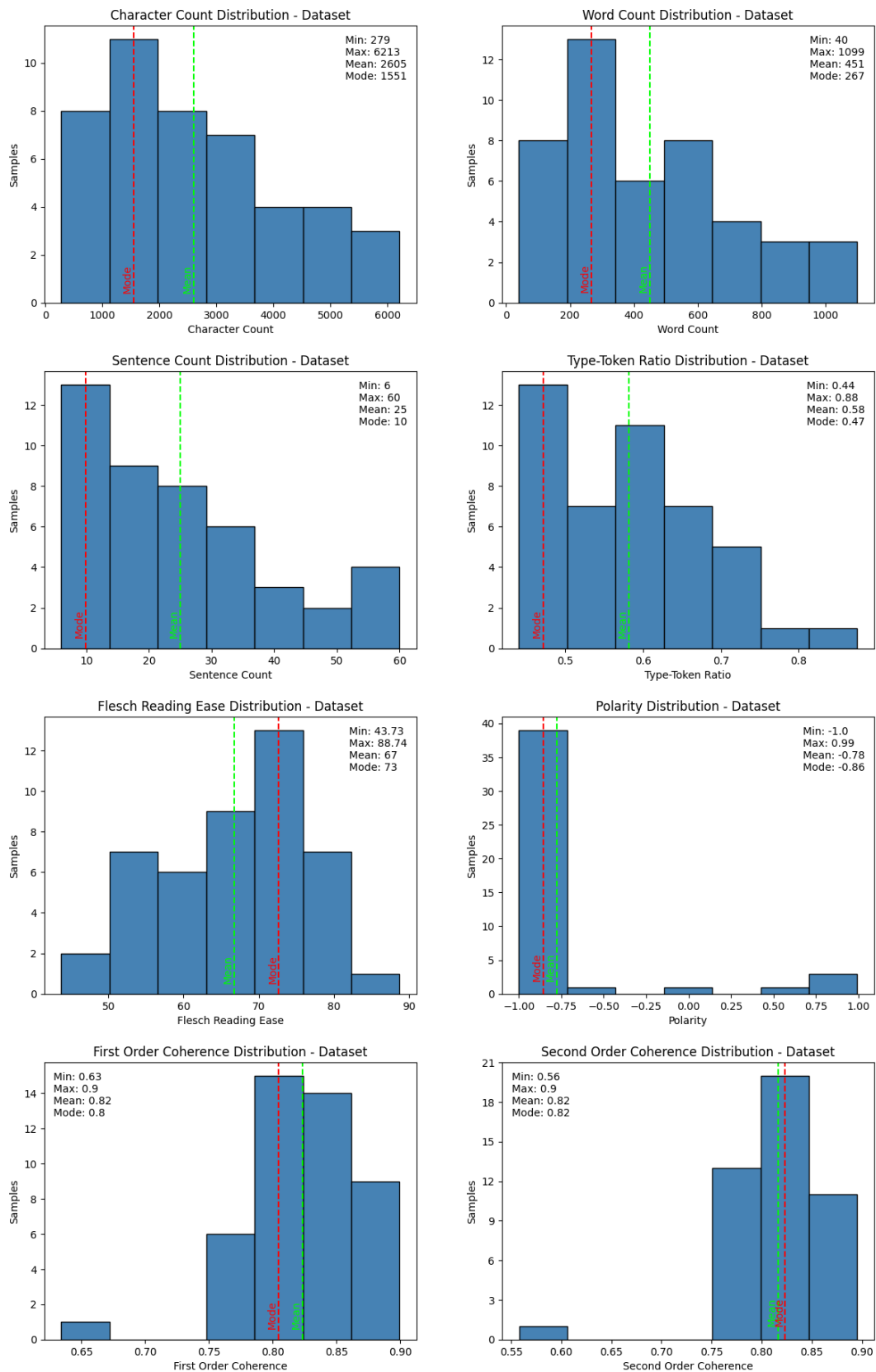


FIGURE 3. Baseline Histogram Distributions of Applicable Metric Scores for Dataset Scenarios.

TABLE 10. BERT Test Scores - Human Evaluation Metrics.

Output	Prompt Adherence	Attacker Success	Techniques Used
1	False	False	N/A
2	False	False	N/A
3	False	False	N/A
4	False	False	N/A
5	False	False	N/A

TABLE 11. BERT Test Scores - LLM-as-a-Judge.

Output	Score	Comments
1	1	Repeated phrases Poor grammar No indication of how prompt is achieved
2	1	Dialogue is unclear No context Collection of non-meaningful phrases and sentences
3	1	Jumbled mess of phrases and sentences No logical conversation or scenario No elements of the prompt
4	1	Jumbled mess of phrases and sentences No logical conversation or scenario No elements of the prompt
5	1	Does not match the prompt Not understandable Collection of unrelated phrases and words

follow the prompt, include a successful attack, or feature social engineering techniques. This reveals that the scenarios generated by BERT are severely low-quality and incomplete.

The LLM-as-a-judge evaluations for the five scenarios generated by BERT are shown in Table 11. The scores of 1 and the strongly negative comments clearly demonstrate that BERT has generated incomprehensible text that cannot be considered as quality scenarios. Although the style metrics provided an indication of low quality, the severely low-caliber results for the human and LLM-as-a-judge evaluations show that style alone is not enough to truly quantify performance. The style metrics suggested that BERT generated easy-to-read and logical text, which is shown to be false.

BERT's lack of inference capabilities caused it to merely regurgitate information from the training dataset and combine parts to form nonsensical outputs. Therefore, even though BERT is suitable for simple NLP tasks, such as summarization, it is unable to perform well at the task of scenario generation, which requires a higher level of reasoning and originality.

E. T5 EVALUATION

The style metric scores for the five scenarios generated by T5 are shown in Tables 12 and 13. Note that the median for each metric is bolded. When compared to the baseline dataset, the scenarios generated by T5 have a similar length, have a much lower level of lexical diversity, are easier to read, have a conversely positive sentiment, are strongly semantically dissimilar, and have a higher degree of logical flow. Overall, these scores are desirable except for the low lexical diversity, high Flesch Reading Ease, positive polarity, and low cosine

TABLE 12. T5 Test Scores - Style Metrics.

Output	Character Count	Word Count	Sentence Count	Type-Token Ratio	Flesch Reading Ease
1	1511	315	12	0.28	83.52
2	1805	338	17	0.27	60.59
3	1458	303	29	0.37	91.37
4	1866	356	21	0.24	93.13
5	1440	310	21	0.28	92.73

TABLE 13. T5 Test Scores - Style Metrics (Cont.).

Output	Polarity	Cosine Similarity	First Order Coherence	Second Order Coherence
1	0.44	0.2	0.94	0.93
2	1	0.14	0.9	0.89
3	0.98	0.18	0.93	0.92
4	-1	0.14	0.9	0.87
5	-0.96	0.17	0.91	0.91

TABLE 14. T5 Test Scores - Human Evaluation Metrics.

Output	Prompt Adherence	Attacker Success	Techniques Used
1	False	False	N/A
2	False	False	N/A
3	False	False	N/A
4	False	False	N/A
5	False	False	N/A

similarity. The scores for these style metrics suggest that T5 created simple scenarios that lack unique content, are overly positive in tone, and vary excessively from the dataset.

The human evaluation metrics for the five scenarios generated by T5 are shown in Table 14. These evaluations are very poor, as T5 failed to generate any scenarios that follow the prompt, include a successful attack, or feature social engineering techniques. This reveals that, as seen with the BERT model, the scenarios generated by T5 are severely low-quality and incomplete.

The LLM-as-a-judge evaluations for the five scenarios generated by T5 are shown in Table 15. The scores of 1 and the strongly negative comments clearly demonstrate that T5 has generated incoherent text that cannot be considered as quality scenarios. As seen with BERT, the style metrics were slightly misleading, as T5 did not produce easy-to-read and logical text. Instead, LLM-as-a-judge indicates that the generated texts lack proper grammar and recognizable narratives.

Although T5 did not regurgitate phrases from the training dataset as seen with the BERT model, it still lacks the inference capabilities and generative power to develop unique text from a short, non-task-prefixed prompt. Therefore, even though T5 achieves good performance with tasks such as question answering and sentiment analysis, it falls behind when instructed to perform feats that are far outside the scope of its pre-training tasks.

TABLE 15. T5 Test Scores - LLM-as-a-Judge.

Output	Score	Comments
1	1	Incoherent mess of phrases and sentences No logic or narrative No understanding of the prompt
2	1	Incoherent and disjointed series of phrases No detail Repeated phrases Lack of proper capitalization, poor grammar
3	1	Incoherent and disjointed series of phrases No clear narrative No context or details No elements of the prompt
4	1	Incoherent and disjointed series of phrases No clear narrative No detail No understanding of the prompt
5	1	Incoherent collection of phrases and sentences Does not form a scenario No clear structure or storyline No connection to the prompt

TABLE 16. GPT-3.5 Test Scores - Style Metrics.

Output	Character Count	Word Count	Sentence Count	Type-Token Ratio	Flesch Reading Ease
1	598	98	7	0.66	79.36
2	907	173	15	0.57	84.27
3	1217	215	13	0.53	63.19
4	2101	374	20	0.48	78.45
5	1117	189	12	0.46	55.44

TABLE 17. GPT-3.5 Test Scores - Style Metrics (Cont.).

Output	Polarity	Cosine Similarity	First Order Coherence	Second Order Coherence
1	-0.86	0.19	0.75	0.75
2	-0.49	0.27	0.66	0.63
3	-0.98	0.35	0.9	0.89
4	-0.91	0.29	0.85	0.85
5	-0.98	0.27	0.91	0.91

F. GPT-3.5 EVALUATION

The style metric scores for the five scenarios generated by GPT-3.5 are shown in Tables 16 and 17. Note that the median for each metric is bolded. When compared to the baseline dataset, the scenarios generated by GPT-3.5 are generally shorter, have a similar level of lexical diversity, are easier to read, have a similarly negative sentiment, are semantically dissimilar, and have a slightly higher degree of logical flow. Overall, these scores are desirable except for the low sentence count and high Flesch Reading Ease. The scores for these style metrics suggest that GPT-3.5 created short and simple scenarios, which indicates an average performance with room for improvement.

The human evaluation metrics for the five scenarios generated by GPT-3.5 are shown in Table 18. These evaluations are all positive, with scores of True for both binary metrics and a good variety of techniques. This indicates that the scenarios generated by GPT-3.5 are relevant, complete, and of good quality.

TABLE 18. GPT-3.5 Test Scores - Human Evaluation Metrics.

Output	Prompt Adherence	Attacker Success	Techniques Used
1	True	True	Pretexting Help Desk Impersonation Phishing
2	True	True	Pretexting Physical Breach Attack Quid Pro Quo Sympathy Urgency
3	True	True	Physical Breach Attack Insider Knowledge
4	True	True	Pretexting Spear Phishing Help Desk Impersonation Malware
5	True	True	Pretexting Insider Knowledge

TABLE 19. GPT-3.5 Test Scores - LLM-as-a-Judge.

Output	Score	Comments
1	3	Easy to understand Format matches the dataset Does not cover all aspects of the prompt Could be improved by adding techniques
2	3	Addresses some aspects of the prompt Format matches the dataset Content is too simple and lacks complexity
3	4	Complete and easy to understand Follows the prompt Format matches the dataset Could be improved by adding more details
4	4	Follows the prompt Simple, coherent, and relevant scenario Format matches the dataset Does not include as much detail as the dataset
5	4	Follows the prompt Complete and easy to understand scenario Format matches the dataset Could be improved by adding more details

The LLM-as-a-judge evaluations for the five scenarios generated by GPT-3.5 are shown in Table 19. The scores of 3 and 4 and the fairly positive comments illustrate that GPT-3.5 has generated satisfactory scenarios of average and good quality. As suggested by the style metrics, LLM-as-a-judge concluded that the generated scenarios are simple and can be expanded upon through complexity, detail, and stronger prompt conformity.

GPT-3.5 was successful in generating quality social engineering scenarios. Unlike BERT and T5, GPT-3.5 possesses the reasoning abilities needed to comprehend the prompt and create a unique story while referencing a baseline dataset. GPT-3.5 also leverages reinforcement learning from human feedback (RLHF) in its creation of especially user-friendly and easy-to-read outputs. However, there is room for improvement, as GPT-3.5 falls behind due to its inability to produce long, detailed scenarios. A more powerful model with stronger reasoning and inference capabilities is required to develop more creative and well-written narratives.

TABLE 20. Llama 3.1: One-Step Test Scores - Style Metrics.

Output	Character Count	Word Count	Sentence Count	Type-Token Ratio	Flesch Reading Ease
1	1174	186	7	0.46	27.56
2	982	182	14	0.42	78.14
3	1734	278	14	0.47	42.82
4	1353	208	10	0.56	24.98
5	1381	238	7	0.47	41.23

TABLE 21. Llama 3.1: One-Step Test Scores - Style Metrics (Cont.).

Output	Polarity	Cosine Similarity	First Order Coherence	Second Order Coherence
1	-0.98	0.24	0.95	0.94
2	-0.54	0.19	0.82	0.83
3	-0.99	0.31	0.93	0.91
4	-0.98	0.29	0.91	0.91
5	-0.99	0.25	0.93	0.94

G. LLAMA 3.1: ONE-STEP EVALUATION

The style metric scores for the five scenarios generated by Llama 3.1: One-Step are shown in Tables 20 and 21. Note that the median for each metric is bolded. When compared to the baseline dataset, the scenarios generated by Llama 3.1: One-Step are generally shorter, have a similar level of lexical diversity, are harder to read, have a similarly negative sentiment, are semantically dissimilar, and have a higher degree of logical flow. Overall, these scores are desirable except for the low sentence count and low Flesch Reading Ease. The scores for these style metrics suggest that Llama 3.1: One-Step created short scenarios that are too difficult for the average reader.

The human evaluation metrics for the five scenarios generated by Llama 3.1: One-Step are shown in Table 22. These evaluations are all positive, with scores of True for both binary metrics and a good variety of techniques. This indicates that the scenarios generated by Llama 3.1: One-Step are relevant, complete, and of good quality.

The LLM-as-a-judge evaluations for the five scenarios generated by Llama 3.1: One-Step are shown in Table 23. The scores of 3, 4, and 5 and the primarily positive comments indicate that Llama 3.1: One-Step has generated suitable scenarios of average, good, and high quality. Despite the low readability suggested by the style evaluation, there is no indication that the generated scenarios are overly complex. In fact, the extra detail makes the scenarios more realistic.

Llama 3.1: One-Step was successful in generating quality prompt-based social engineering scenarios. Llama 3.1 possesses the complex reasoning abilities needed to comprehend the prompt and create a unique and detailed story while referencing a baseline dataset. Llama 3.1 also makes use of grouped-query attention in its strong inference capabilities, which aids in its production of creative and well-written scenarios. As a result, Llama 3.1: One-Step goes beyond GPT-3.5 in the quality of its generated scenarios. However, there is room for improvement, because some texts generated

TABLE 22. Llama 3.1: One-Step Test Scores - Human Evaluation Metrics.

Output	Prompt Adherence	Attacker Success	Techniques Used
1	True	True	Pretexting Spear Phishing Help Desk Impersonation Business Email Compromise
2	True	True	Pretexting Help Desk Impersonation Physical Breach Attack
3	True	True	Pretexting Spear Phishing Help Desk Impersonation Quid Pro Quo Urgency
4	True	True	Insider Knowledge Password Attack
5	True	True	Pretexting Spear Phishing Help Desk Impersonation Business Email Compromise

TABLE 23. Llama 3.1: One-Step Test Scores - LLM-as-a-Judge.

Output	Score	Comments
1	4	Clear and detailed scenario Follows the prompt Includes actions before, during, and after the attack Format matches the dataset Could be improved with more specific details
2	3	Comprehensible Addresses aspects of the prompt Format matches the dataset Short and less detailed
3	4	Format does not match the dataset Detailed scenario Covers all aspects of the prompt Includes consequences of the attack Could be improved with more specific details
4	4	Complete and easy to understand Follows the prompt Format matches the dataset
5	5	Detailed and well-crafted scenario Covers all aspects of the prompt Format matches the dataset Provides background info

by Llama 3.1: One-Step lack the length and detail of a truly excellent scenario. This may be achievable by using interactive prompting instead to guide Llama 3.1 to develop a more in-depth scenario step by step.

H. LLAMA 3.1: INTERACTIVE EVALUATION

The style metric scores for the five scenarios generated by Llama 3.1: Interactive are shown in Tables 24 and 25. Note that the median for each metric is bolded. When compared to the baseline dataset, the scenarios generated by Llama 3.1: Interactive have a similar length, have a slightly lower level of lexical diversity, are harder to read, have a similarly negative sentiment, are semantically dissimilar, and have a higher degree of logical flow. Overall, these scores are desirable except for the slightly low type-token ratio and low Flesch Reading Ease. The scores for these style metrics suggest that Llama 3.1: Interactive created scenarios that

TABLE 24. Llama 3.1: Interactive Test Scores - Style Metrics.

Output	Character Count	Word Count	Sentence Count	Type-Token Ratio	Flesch Reading Ease
1	1257	212	11	0.48	51.89
2	2866	485	25	0.42	52.49
3	3533	600	42	0.4	65.42
4	2487	414	21	0.45	51.48
5	2157	357	17	0.51	51.38

TABLE 25. Llama 3.1: Interactive Test Scores - Style Metrics (Cont.).

Output	Polarity	Cosine Similarity	First Order Coherence	Second Order Coherence
1	-0.98	0.31	0.93	0.91
2	0.96	0.22	0.91	0.9
3	1	0.31	0.82	0.83
4	-0.93	0.32	0.92	0.92
5	-0.99	0.25	0.91	0.9

TABLE 26. Llama 3.1: Interactive Test Scores - Human Evaluation Metrics.

Output	Prompt Adherence	Attacker Success	Techniques Used
1	True	True	Pretexting Spear Phishing Phishing Business Email Compromise Insider Knowledge
2	True	True	Pretexting Spear Phishing Physical Breach Attack Insider Knowledge
3	True	True	Pretexting Spear Phishing Sympathy
4	True	True	Pretexting Spear Phishing Help Desk Impersonation Insider Knowledge Urgency Malware
5	True	True	Pretexting Spear Phishing Website Spoofing Urgency

somewhat reuse words and are too difficult for the average reader. Additionally, although the median polarity score is strongly negative, two outputs received strongly positive polarity scores. This dichotomy may have occurred due to the scenario's focus on the attacker's success, or a superficially positive interaction between the attacker and victim.

The human evaluation metrics for the five scenarios generated by Llama 3.1: Interactive are shown in Table 26. These evaluations are all positive, with scores of True for both binary metrics and a good variety of techniques. This reveals that the scenarios generated by Llama 3.1: Interactive are relevant, complete, and of good quality.

The LLM-as-a-judge evaluations for the five scenarios generated by Llama 3.1: Interactive are shown in Table 27. The scores of 4 and 5 and the very positive comments demonstrate that Llama 3.1: Interactive has generated strong

TABLE 27. Llama 3.1: Interactive Test Scores - LLM-as-a-Judge.

Output	Score	Comments
1	4	Clear and detailed scenario Follows the prompt Easy to understand Format matches the dataset Could be improved with more details and techniques
2	5	Detailed and well-crafted Covers all aspects of the prompt Uses several techniques Format matches the dataset
3	5	Very detailed and convincing story Covers all aspects of the prompt Format matches the dataset Could be shortened for clarity
4	5	Excellent scenario that creates a story Detailed and well-crafted Covers all aspects of the prompt Format matches the dataset Uses multiple social engineering techniques Uses realistic elements
5	5	Excellent scenario that creates a story Detailed and well-crafted Covers all aspects of the prompt Format matches the dataset Includes attack planning, execution, and response

scenarios of good and high quality. Although the style metrics suggested low lexical diversity and low readability, there is no indication that the generated scenarios are excessively repetitive or complex. Instead, the extra detail, multi-step attacks, and in-depth language make the scenarios more compelling, story-like, and similar to real-life attacks.

Llama 3.1: Interactive was successful in generating high-quality prompt-based social engineering scenarios. As demonstrated by the One-Step model version, Llama 3.1 possesses the complex reasoning and strong inference capabilities needed to comprehend the prompt and create a unique, creative, and very detailed story while referencing a baseline dataset. With its interactive prompt model, Llama 3.1: Interactive goes beyond the One-Step version to create longer and more involved scenarios that are highly customized to user specifications. The step-by-step nature of the generation process enables Llama 3.1 to more thoroughly reason about each stage of the scenario as it is developed, thereby resulting in more realism and depth. When compared to the other four model configurations, Llama 3.1: Interactive generates the most excellent scenarios.

I. COMPARATIVE EVALUATION

The style metric scores, human evaluation metric scores, and LLM-as-a-judge metric scores for the five scenarios generated by each model configuration are visualized using bar plots in Figs. 4, 5, 6, and 7. Note that the 'techniques used' style metric and the LLM-as-a-judge comments are omitted, as they are non-numeric. These visualizations enable quick performance comparisons among each model configuration and between each configuration and the dataset. Based on our comprehensive cross-metric evaluation, it is clear that the Llama 3.1: Interactive model achieved the best overall performance. Through metrics, Llama 3.1: Interactive

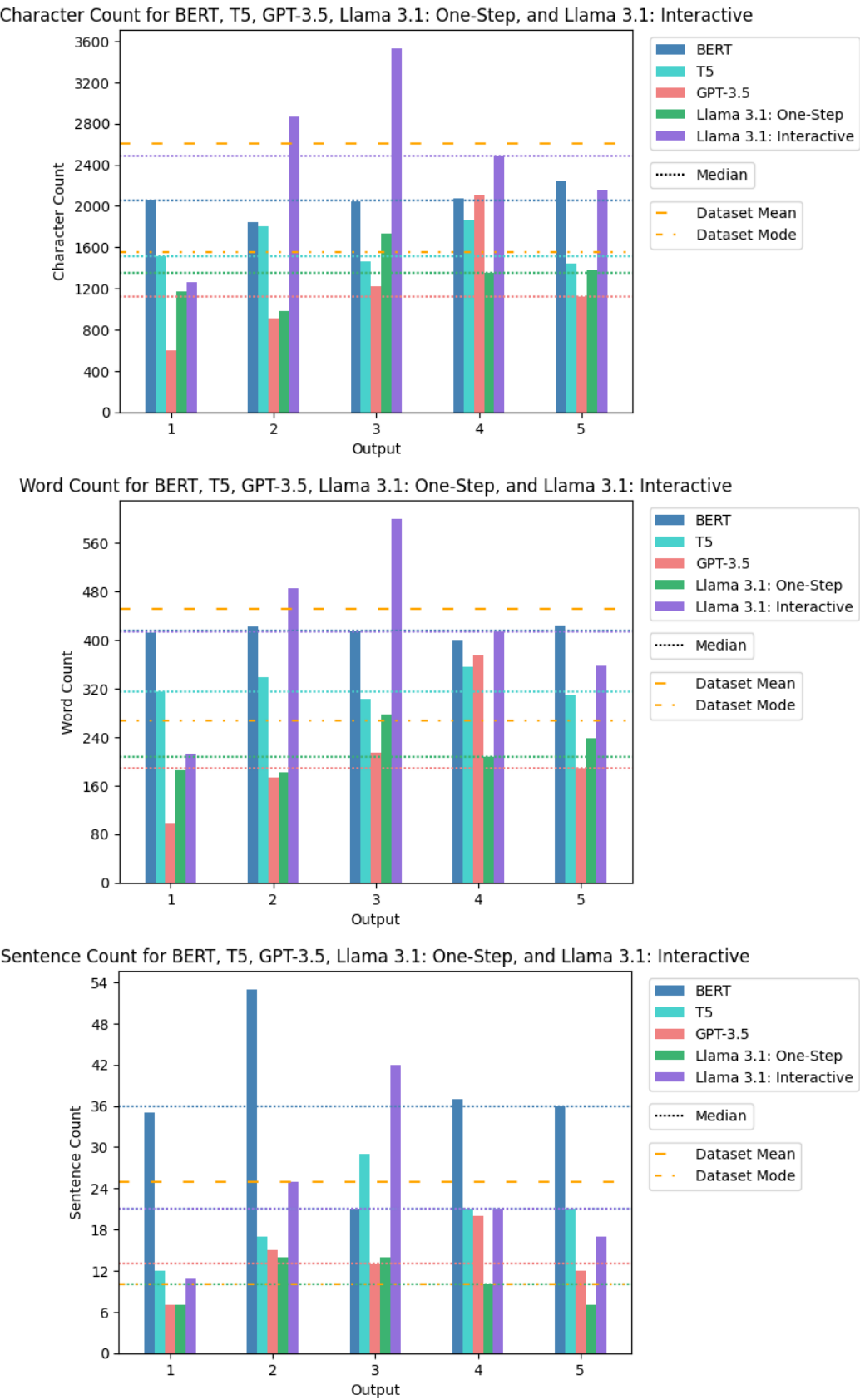


FIGURE 4. Style Metric Scores for BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive.

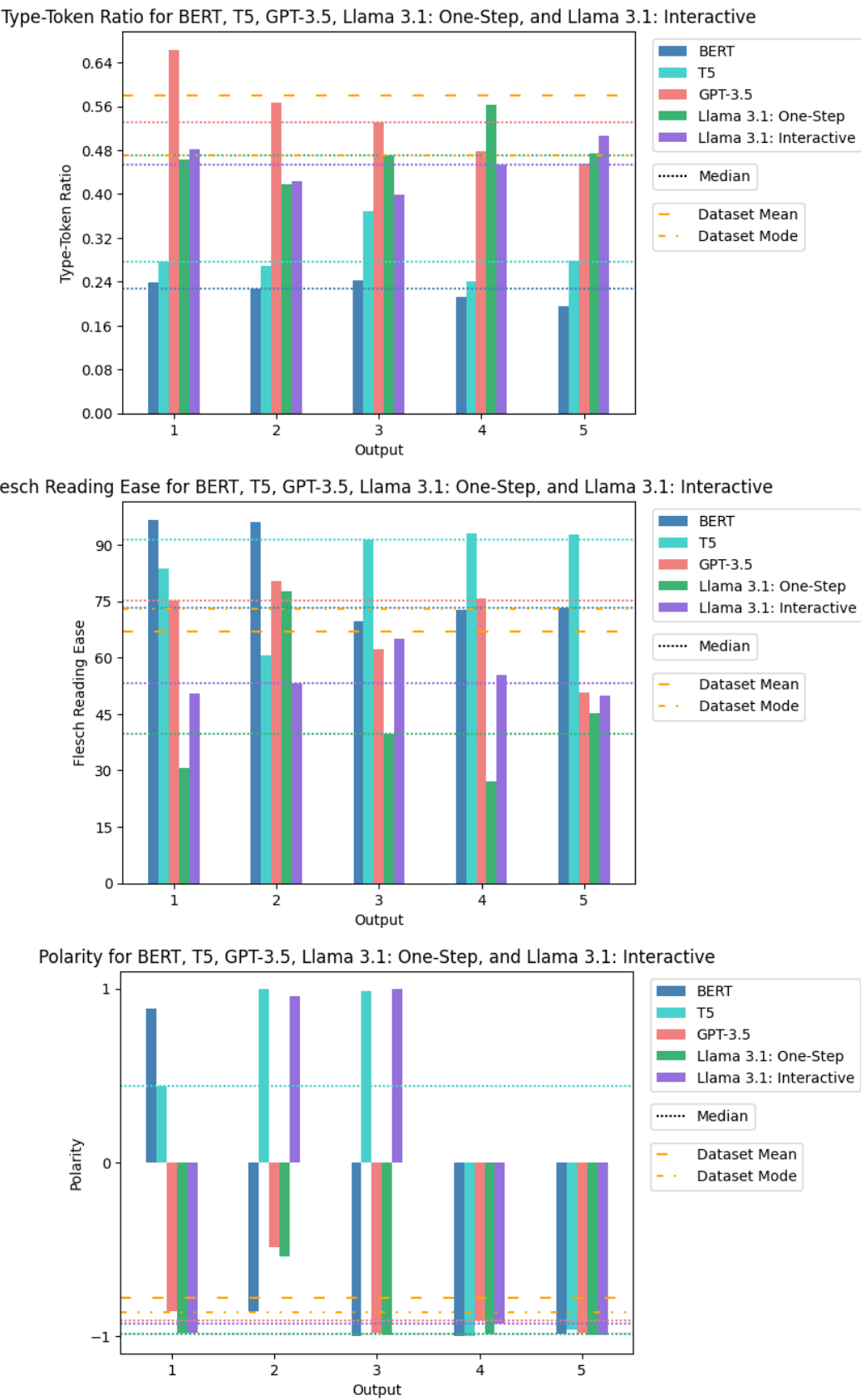


FIGURE 5. Style Metric Scores for BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive (Cont.).

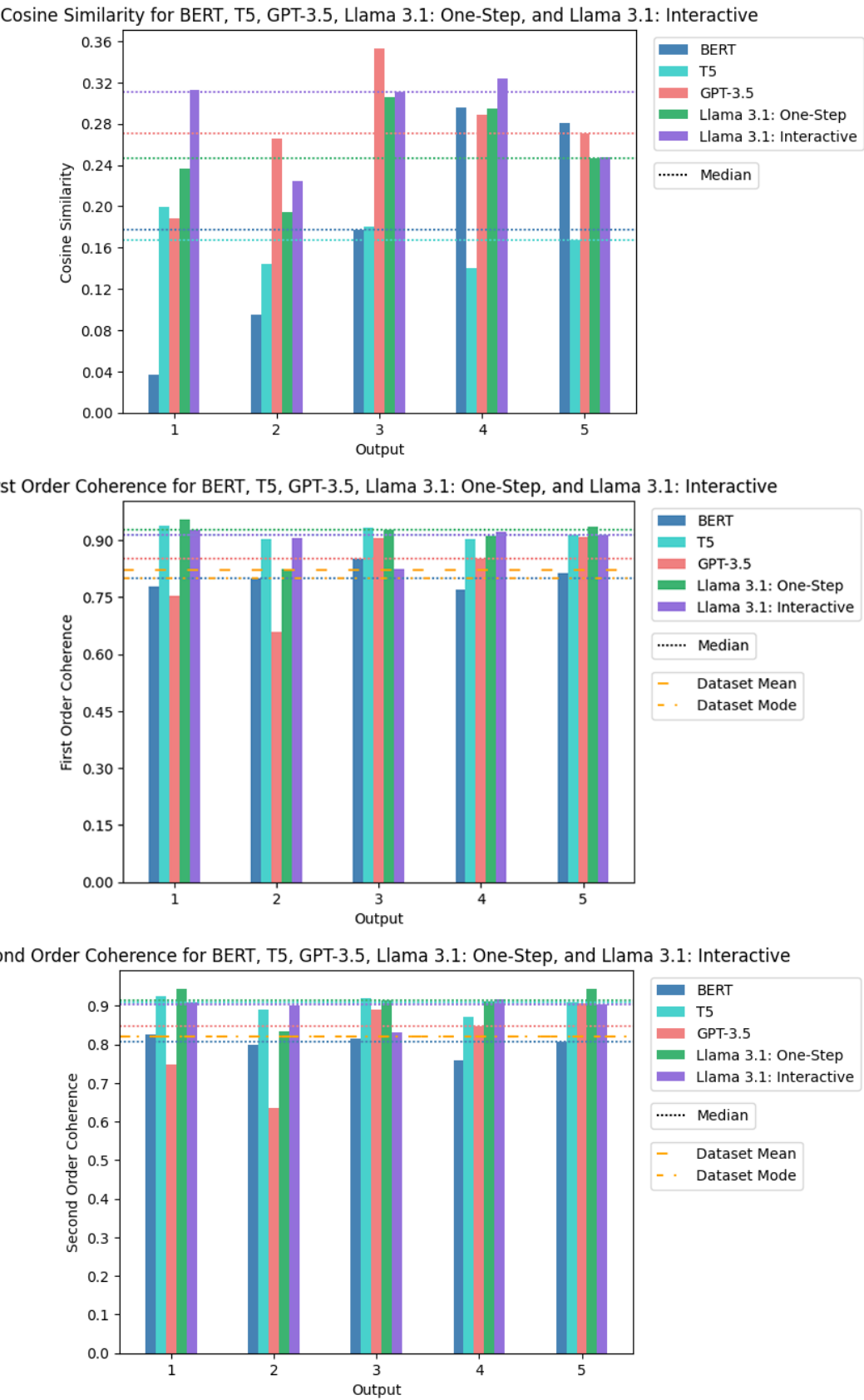


FIGURE 6. Style Metric Scores for BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive (Cont.).

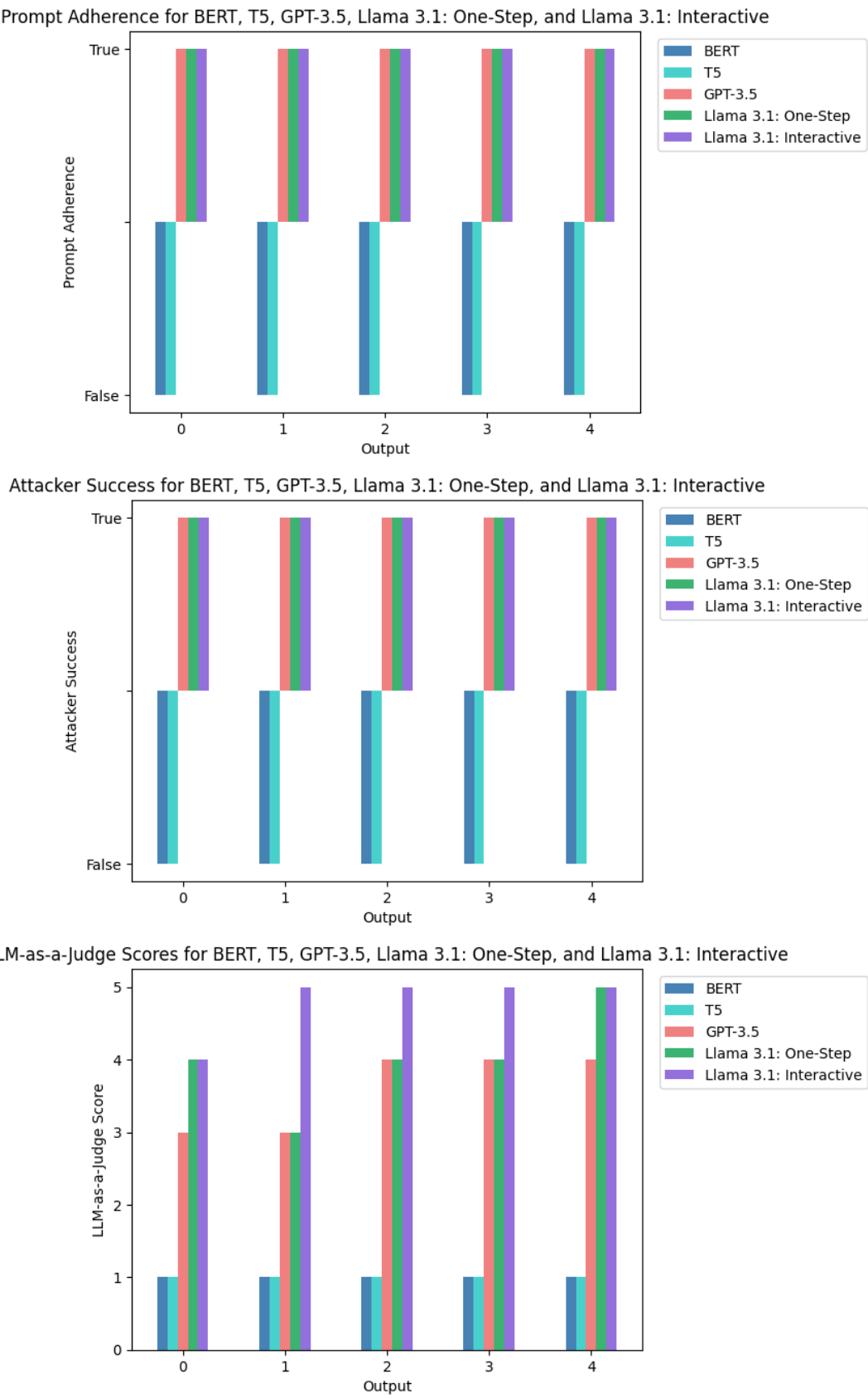


FIGURE 7. Human Evaluation and LLM-as-a-Judge Scores for BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive.

demonstrated an ability to produce high-quality social engineering scenarios with strong realism, relevance, clarity, detail, depth, coherence, writing quality, and applicability.

J. LIMITATIONS

Although Llama 3.1: Interactive was the model configuration that produced the highest quality scenarios, its performance is limited. The selected model, Llama 3.1 8B Instruct, is the most lightweight version of Llama 3.1. The 70B and 405B versions, with 70 billion and 405 billion respective parameters, are drastically more powerful and would likely produce more rich scenarios. Beyond the scope of Llama 3.1, state-of-the-art models such as DeepSeek-R1, GPT-4o, and the new Llama 4 may be utilized to produce highly intricate scenarios. Additionally, in this experiment, Llama 3.1 was only tuned using prompt engineering and hyperparameter tuning. Therefore, instead of simply referencing the dataset in the prompt, adding a step for fine-tuning on the dataset may improve performance. In their current form, even the most powerful LLMs are unable to fully replicate the creativity, reasoning, and contextual understanding of humans, yet they are still incredibly useful for automated generation tasks.

The dataset used in this study is small (only 45 samples) and outdated (published in 2002). Many of the scenarios in the dataset use non-contemporary language and lack social engineering techniques and trends of the present day. It is hard to collect real scenario examples since most attacks contain sensitive information, which is why this fictional dataset was the best currently available option to the authors' knowledge. However, a larger, more modern dataset of attack scenarios is ideal and may contribute to more robust models.

K. APPLICATIONS

1) CYBERSECURITY AWARENESS

The scenarios generated by Llama 3.1: Interactive are of sufficiently high quality to supplement improved cybersecurity awareness training programs. Our analysis supports that Llama 3.1: Interactive generates realistic, contextually appropriate, detailed, and well-written scenarios that include the use of multiple relevant attack methods by a savvy and successful attacker. As discussed in Section II-D, many currently deployed curriculums are unable to affect strong awareness skills in their students, partially due to the lack of quality hands-on training. Most do not involve attack simulations, and those that do incorporate them emphasize phishing, yet ignore the multitude of other potent social engineering techniques and the complexity of most contemporary attacks. Although Verizon and IBM assert that awareness is the important factor in protecting against attacks and breaches, resilience dwindles without impactful training. This issue highlights the opportunity for our Llama 3.1: Interactive model to be leveraged to integrate realistic synthetic social engineering scenarios into cybersecurity awareness programs to support more effective learning among participants.

In addition, sourcing sample scenarios for training programs is difficult, as raw examples of real attacks are inaccessible due to security restrictions, and manually hand-writing complex scenarios is difficult and costly. Our Llama 3.1: Interactive model solves this issue by providing an avenue for the efficient guided development of custom scenarios to fit the needs of any situation. This enhancement has a strong potential to improve the public's ability to recognize a variety of unique and in-depth attacks, with the goal of building resilience to future attacks through awareness.

There are multiple training use-cases for synthetic scenarios generated by Llama 3.1: Interactive. The primary application is software-based attack simulations, in which users can observe or interact with an artificial attacker in a realistic environment. This provides practical and impactful hands-on training to introduce learners to common attack phases, types, and strategic language, with the goal of developing solid attack recognition and mitigation skills. In the realm of cybersecurity awareness, accessibility is a prevalent issue, as most employer-led programs lack simulation training and there are no widely adopted at-home learning resources for the general public. Our model allows for efficient development of realistic scenarios, which can be integrated into both work-based programs and simple at-home learning, to support access to high-quality training for all. Due to the customizable nature of Llama 3.1: Interactive, it can be utilized to produce social engineering scenarios that can be used in specialized training simulations, such as attacks that are specific to a company, job title, or architecture. Another potential use case is the development of threats that incorporate specific attack strategies that are trending, novel, or particularly dangerous. In addition, simulations can be adjusted for groups of different ages and levels of technology experience. Apart from software-based simulations, generated scenarios can be applied to role-playing workshops, to guide learning in a face-to-face setting with opportunities for questions and commentary. Synthetic scenarios can also be used to audit existing training programs to indicate effectiveness, identify students' areas of weakness, and discover ways to improve curriculum. Fake attacks can be sent to trainees, whether they are employees or program subscribers, and data can be collected regarding message interaction, link click-through, and attack reporting rates.

2) OTHER APPLICATIONS

Generated scenarios can also be used to supplement attack detection software. Due to the challenges of collecting or obtaining raw attack samples, a synthetic dataset of realistic social engineering scenarios is highly valuable. Such a dataset can be used to improve existing and in-development detection programs, and can be particularly useful for training and evaluating AI models.

Red teaming is another avenue in which generated scenarios can be applied. Red teams can utilize Llama 3.1:

Interactive to develop simulations involving novel or edge case attacks to evaluate their company's defenses and discover subtle vulnerabilities. Our model is capable of assisting red teams in terms of efficiency and scenario creativity, which supports overall productivity and proactivity.

The scenario generation models developed in this study may also be misused, as criminals can utilize these tools to develop attacks and deploy them against the public. Although the customizability of Llama 3.1: Interactive is beneficial for creating specific training simulations, it can be exploited to generate highly targeted attack content. This raises questions about whether it can be guaranteed that these models are used solely for ethical applications.

IX. CONCLUSION

The following section will describe the challenges faced during this study, provide a comprehensive overview of our findings, and outline possible future work on this topic.

A. CHALLENGES

Several challenges were faced throughout the duration of this study. The absence of existing works that involve the topic of social engineering scenario generation or involve data sourced from *The Art of Deception* [17] made it more challenging to devise experiments and left the authors without solid references to compare our results with.

The authors originally intended to use a variant of GPT-4 for the second experiment, but a roadblock was encountered during fine-tuning. During our attempts to train GPT-4o and GPT-4o mini, OpenAI blocked the fine-tuning process from completion due to a moderation flag. The malicious nature of the dataset scenarios violated OpenAI's usage policies for those models, yet it was able to be successfully used to fine-tune GPT-3.5.

The prompt engineering process was highly demanding, as the Llama 3.1 model's behavior was often unpredictable. This made it very difficult to craft a solid prompt that resulted in outputs that aligned with the instructions. One major issue was that Llama 3.1 would sometimes ignore parts of the prompt or simply repeat the prompt as output. Also, due to the literal nature of LLMs, it was difficult to describe certain nuanced instructions. One example is the trade-off between the skills required for the attacker to achieve their goal and the victim's level of suspicion as well as their willingness to cooperate. It was difficult to guide Llama 3.1 to generate a scenario in which the attacker must effectively use multiple strategies in order to deceive the victim, and the victim is not fooled by a blatant attack but can still be ultimately manipulated. Additionally, even after tuning the hyperparameters related to frequency and sampling, Llama 3.1 would occasionally produce outputs that got stuck in a repetitive loop midway through the text. Similarly, even after adjusting the hyperparameters that controlled model creativity and token sampling, Llama 3.1 still produced deterministic outputs in some cases.

Since the task of scenario generation relies on uniqueness, there were no targets to compare the model outputs against. This made finding applicable metrics very difficult, since common standards such as accuracy, F1, ROUGE, and BLEU depend on a reference output. As a result, a mix of style metrics, human evaluation metrics, and LLM-as-a-judge was used.

B. FINDINGS

This study illustrated the manipulation of LLMs of varying architectures: BERT, T5, GPT-3.5, and Llama 3.1, with the goal of generating high quality social engineering scenarios. A small, manually collected dataset of sample attack scenarios was used to supplement these models to aid in their creation of unique scenarios that incorporate similar social engineering techniques. Two techniques were used to refine the selected LLMs for this domain-specific task: fine-tuning (BERT, T5, and GPT-3.5) and prompt engineering (Llama 3.1).

Five models (BERT, T5, GPT-3.5, Llama 3.1: One-Step, and Llama 3.1: Interactive) were developed, as Llama 3.1 was used to generate scenarios in both single-step and incremental processes. After refinement and hyperparameter tuning, the configuration with the optimal results for each model was evaluated on a test set of five instruction prompts. Each output scenario was evaluated using a comprehensive series of criteria, namely style metrics, human evaluation metrics, and LLM-as-a-judge. Llama 3.1: Interactive was the strongest model, as it produced high-scoring social engineering scenarios of superior quality that are viable for use in real-life simulation training curriculums. Llama 3.1: One-Step was the second best model, as it fell behind only in scenario detail. GPT-3.5 was able to produce average-level scenarios, yet the model suffered from output simplicity. Lastly, T5 and BERT performed poorly, as they produced incoherent segments of incomprehensible text.

These performance differences can be attributed to the varying capabilities of each model. Llama 3.1 is the strongest model due to its very high parameter count (8 billion), extremely large pre-training dataset (15 trillion tokens), and support for conversation, contextual understanding, complex reasoning, and inference. It rises above GPT-3.5, which has a smaller architecture, a smaller pre-training dataset, and boasts less powerful reasoning abilities. T5 and BERT are the least powerful models, as they contain more basic architectures that are better suited for more straightforward NLP applications. Llama 3.1 and GPT-3.5 are also both decoder-only architectures, compared to the encoder-decoder T5 and the encoder-only BERT. This confirms the strength of robust decoder-only architectures for inference-heavy autoregressive tasks.

C. FUTURE WORK

This study may be expanded with the aim of creating models capable of producing even higher-quality social engineering scenarios. The primary domain for improvement

is experimentation using more powerful LLMs, as mentioned in Section VIII-J. However, there is more to include beyond the use of different models.

One area for exploration is the previously mentioned RLHF and its counterpart, reinforcement learning from AI feedback (RLAIF) [52]. RLHF helps models to learn to produce more user-friendly outputs by leveraging human feedback, whereas RLAIF automates this process by using feedback from an LLM, which solves RLHF scalability issues [52]. Even though some models such as GPT-3.5 are pre-trained using RLHF, most LLMs can be further refined by incorporating customized RLHF or RLAIF during tuning.

Another area for exploration is retrieval-augmented generation (RAG) [53]. RAG supplements LLMs by providing access to additional external knowledge bases without the need for retraining [53]. This method may be beneficial for this application, as it would allow an LLM to retrieve and reference documents that describe current social engineering techniques and trends which are outside the knowledge base of the pre-training and fine-tuning datasets.

In the future, the authors would like to experiment with the application of RLHF, RLAIF, and RAG to LLM-based social engineering scenario generation, and to compare the results to the outcomes of this study.

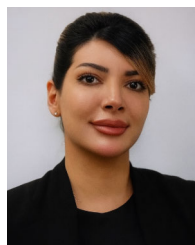
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